
Exercise 9

1 Nonlinear MPC

The following is a dynamic model of a nonlinear double integrator:

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = f(x, u) = \begin{bmatrix} u \\ \alpha(x_1) + x_1 \end{bmatrix}$$

subject to constraints $\|x\|_\infty \leq 1$ and $|u| \leq 1$. Treat α as a symbolic function.

1. Discretize the system $x^+ = \phi(x, u)$ with a sample period of $h = 1$ seconds using Runge-Kutta 2. Assume that during the sample period, the input is constant, i.e., $u(t) = u_i$ for all $t \in [h(i), h(i+1)]$ for $i \in \{0, 1, 2, \dots\}$.
2. Linearize the discrete-time system around the point $x_s = (0, 0)$ and $u_s = 0$ and for $\alpha(x_1) = \sin(x_1)$.
3. Design a stabilizing and recursively feasible linear MPC controller for the linear discrete-time model computed in the previous part.
4. If the input from your linear MPC controller is applied to the nonlinear system, will the closed-loop system necessarily satisfy the constraints?
 - If yes, prove that this is the case.
 - If no, give suggestions for how you can modify your controller to ensure that it does.